

1. (a) State the definition for a set in $\mathbb{R}$ to be closed.

(b) State the definition of a function, $f : A \rightarrow \mathbb{R}$, is continuous on $A \subset \mathbb{R}$.

(c) Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be two continuous functions. Prove that the set E is closed, where

$$E = \{x \in \mathbb{R}, f(x) = g(x)\}$$

2. (a) State the definition of the closure $\overline{A}$ of a set A .

(b) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function, and let $A \subset \mathbb{R}$. Prove that $f(\overline{A}) \subset \overline{f(A)}$.

(c) Determine whether that $f(\overline{A}) = \overline{f(A)}$ always holds (either provide a proof, or construct an example to back up your answer).

3. (a) State the definition of a compact set, and the Heine-Borel theorem.

(b) Let $\{K_\lambda, \lambda \in \Lambda\}$ be a collection of compact sets i.e., each K_λ is compact. Prove that $K = \bigcap_{\lambda \in \Lambda} K_\lambda$ is compact.

(c) Assume that K in (b) contains no limit point. Prove that K is a finite set.

4. (a) State the definition that $f : A \rightarrow \mathbb{R}$ is uniformly continuous.

(b) Let $f : A \rightarrow \mathbb{R}$ be uniformly continuous. Let x be a limit point of A i.e., $x \neq x_N \in A, x_n \rightarrow x$. Prove that the limit, $\lim_{n \rightarrow \infty} f(x_n) = L$, for some $L \in \mathbb{R}$.

(c) Let $f : [2, \infty) \rightarrow \mathbb{R}, f(x) = \sqrt{x-2}$. Prove that $f(x)$ is uniformly continuous.

5. (a) State the sequential definition for a function, $f : A \rightarrow \mathbb{R}$, to be continuous on A .

(b) Let $f : (a, b] \rightarrow \mathbb{R}$ and $g : [b, c) \rightarrow \mathbb{R}$ be continuous functions. Assume that $f(b) = g(b)$. Prove that $h : (a, c) \rightarrow \mathbb{R}$ is a continuous function, where $h(x)$ is defined by

$$h(x) = \begin{cases} f(x), & x \in (a, b] \\ g(x), & x \in [b, c) \end{cases}$$

6. Determine whether the following statements are true (**T**) or false (**F**); if choosing **T**, must provide with a reason or the statement of a theorem, if choosing **F**, you must provide with a counterexample
- (a) Let $f : A \rightarrow \mathbb{R}$ be a function. If f is continuous and $x_n \in A$ is a Cauchy sequence, then $f(x_n)$ is also a Cauchy sequence
 - (b) Let $f : A \rightarrow \mathbb{R}$ be a continuous function. If U is open, then $f^{-1}(U)$ is open, where $f^{-1}(U) = \{x \in A, f(x) \in U\}$.
 - (c) If A is a perfect set, then A is also a connected set.
 - (d) If K is a compact set, A is a closed set, then $K \cap A$ is a compact set.
 - (e) Let $f : \mathbb{R} \rightarrow \mathbb{R}$. If there is a dense subset $A \subset \mathbb{R}$ such that f is continuous on A , then f is continuous on $\mathbb{R}$.
 - (f) Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous function. Then there is a $c \in [0, 1]$ such that $f(c) = c$.